

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT3 — VISUALIZATION CRITIQUE & REDESIGN

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Critique & Redesign
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Evaluate flawed or misleading data visualizations, critique their specific shortcomings, and justify an improved redesign.(BL5)</i>		
Student Name:	Kola Amrutha Sandhya	Reg. No.:	192524270
Section / Batch:	AIDS	Date:	18-08-2026

Code of Conduct

I Kola Amrutha Sandhya 192524270 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond simply naming a flaw: explain WHY it is misleading and what a reader could wrongly conclude from it.
- Your redesign must be a specific, justified chart proposal — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION CRITIQUE & REDESIGN

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT3 — Visualization Critique & Redesign
CO3 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO3-AT3 Visualization Critique & Redesign problems, in which students are given a flawed or misleading chart, must critique its specific shortcomings, and propose a justified redesign.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the specific visualization principle being violated (e.g., proportional ink, axis scaling, color perception, overplotting) in the given flawed chart.	Good understanding of the relevant principle, with only minor gaps in identifying exactly what is violated.	Basic understanding, but with some misconceptions about which principle applies to the flaw shown.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	25%	Clearly explains why the flaw is misleading, including what a reader could wrongly conclude as a result — not just that something is 'wrong.'	Explains the flaw with adequate reasoning, covering most of why it misleads a reader.	Identifies the flaw but explanation is generic or only loosely tied to its misleading effect.	Little or no explanation of why the flaw is misleading; the critique is a bare assertion.
Application	20%	Proposes a specific, well-justified redesign that directly resolves the identified flaw,	Proposes a workable redesign that resolves the flaw, with only minor gaps in	Proposes a redesign, but it only partially resolves the flaw or lacks sufficient justification.	Fails to propose a workable redesign, or the redesign does not address the identified flaw.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		correctly applying appropriate chart-selection and design principles.	justification.		
Organization	15%	The response is clearly structured, addressing both the critique and the redesign in a logical sequence with coherent overall flow.	The response is organized and addresses both parts, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the critique and the redesign are completely accurate and well-suited to the specific chart and data described.	Mostly accurate, with only minor omissions or a small error in the critique or redesign.	Partially correct; the critique or redesign contains notable inaccuracies or mismatches to the scenario.	Largely incorrect or inappropriate, with major gaps in both the critique and the redesign.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

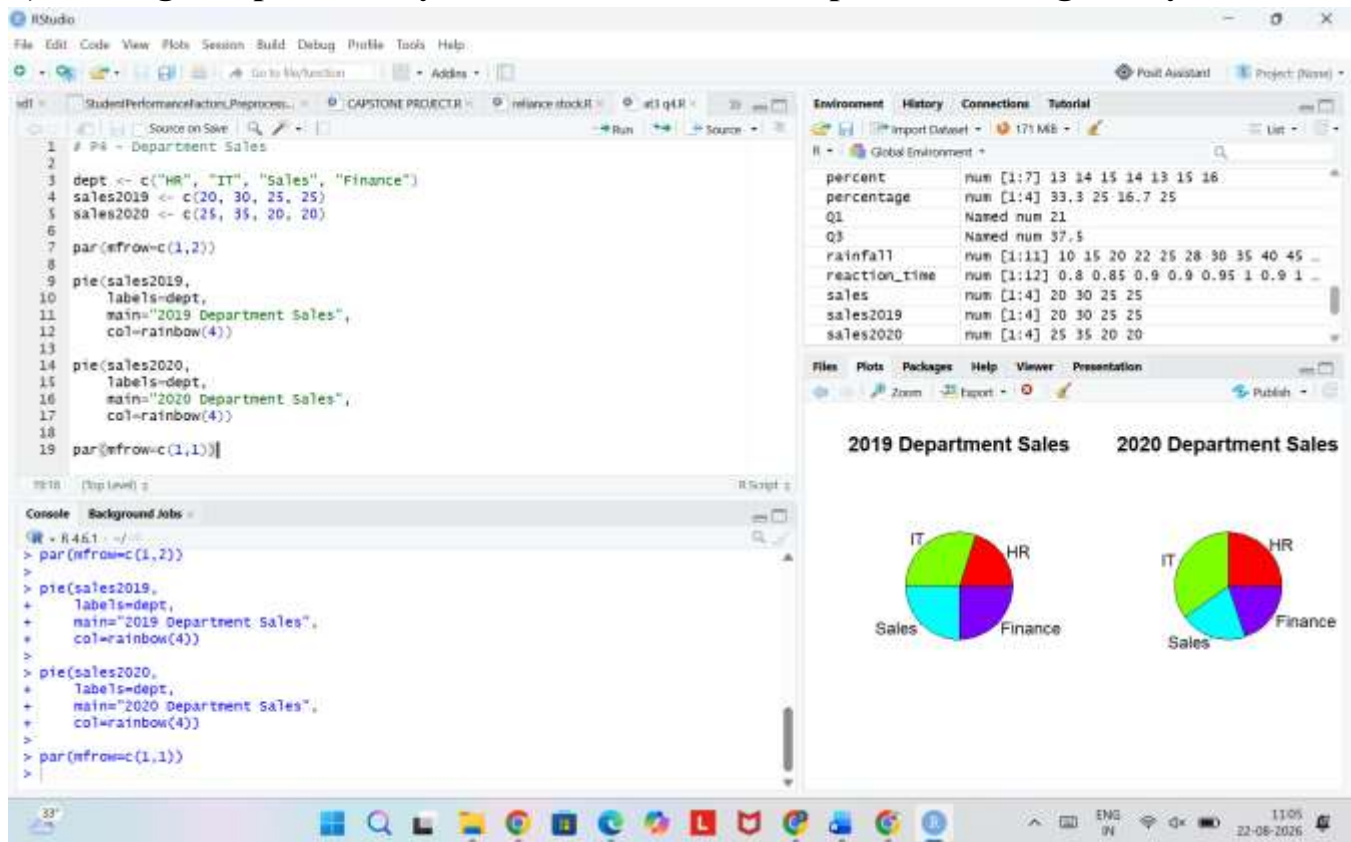
Problem 4

Scenario: A company's 2020 pie chart of sales by department orders the slices randomly around the circle, while an earlier 2019 chart for the same departments used alphabetical order.

circle, while an earlier 2019 chart for the same departments used alphabetical order.

a) Critique the inconsistency between the two charts and its effect on year-over-year comparison.

b) Redesign: explain how you would fix this for a report containing both years.

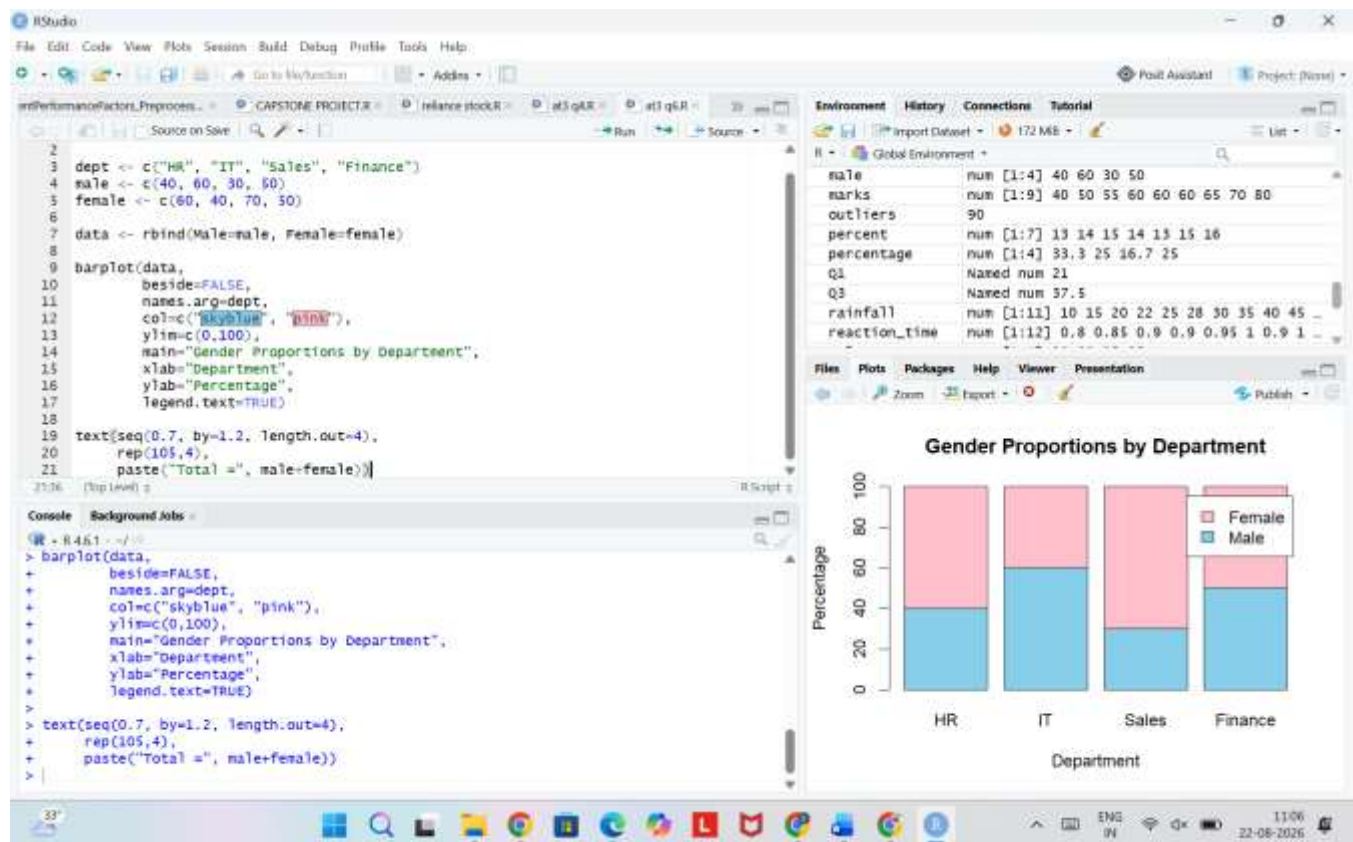


Problem 6

Scenario: A report converts department gender data into a 100% stacked bar chart without showing the total headcount of each department.

a) Critique what critical information is hidden by this design.

b) Redesign: propose how to present the data so the hidden information becomes visible.

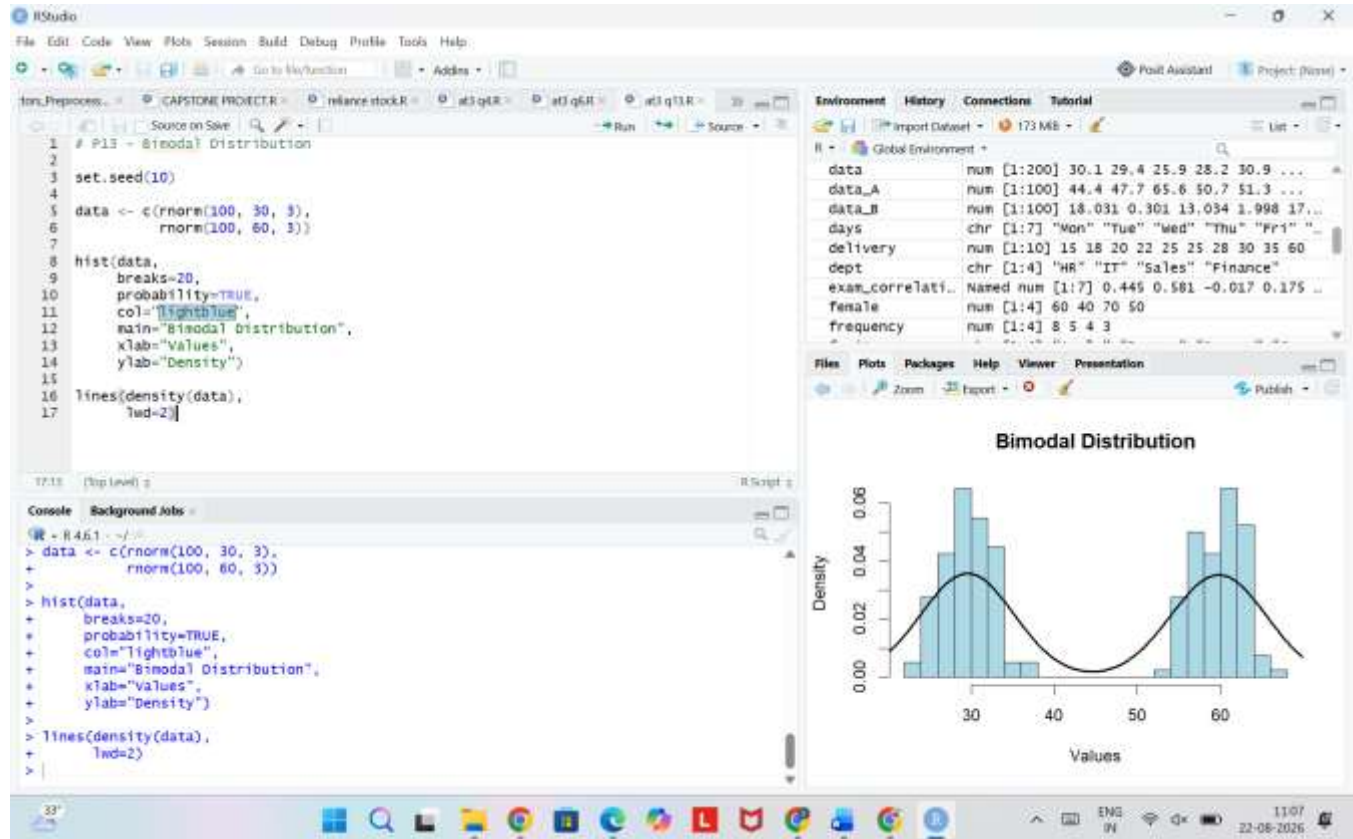


Problem 13

Scenario: A box plot is used to compare two datasets that are later revealed to be bimodal (two distinct peaks), but the box plot shows no visible sign of this.

a) Critique what limitation of box plots this scenario reveals.

b) Redesign: propose an alternative chart that would reveal the bimodal pattern.



Problem 20

Scenario: A manager compares two employees' performance using two separate pie charts with very different total values, but draws both circles at the same visual size on the page.

a) Critique the misleading impression created by drawing both circles the same size.

b) Redesign: propose a fairer way to compare the two employees.

